Yeast spp

Yeasts have been identified which exhibit probiotic characteristics. With our vast knowledge of yeasts, particularly their biology, genetics and safety issues, their potential as probiotics is good.

- Yeast are unicellular fungi
- Dominate fungal diversity in the oceans
- Large cells 20-300 times larger than bacteria
- Yeasts are not affected by antibiotics.
- Cell wall structure comprise β-glucans
- glucans is used as immnuo stimulants
- Yeast is used in the backing industry and for producing alcoholic beverages
- Yeast are used in ruminant feeding to reduce oxygen level in the rumen
- Natural occurrence in healthy fish
- Debarryomyces hansenii is a "salt loving" yeast as Sodium is not toxic

This is advantageous in probiotic preparations used for preventing disturbances in the normal microflora in the presence of antibacterial metabolites.

Saccharomyces Cerevisiae :

"Saccharomyces" derives from Latinized Greek and means "sugar mold" or "sugar fungus", Saccharo- being the combining form "sugar-" and myces being "fungus". Cerevisiae comes from Latin and means "of beer". S. cerevisiae is a species of budding yeast. It is perhaps the most useful yeast owing to its use since ancient times in baking and brewing.

Natural fermentation product made of 100% S. cerevisiae cells, which are grown in sugar cane molasses, having a significant importance in animal feeding, not only because of its high nutritional value, but also because of its known prophylactic effect. Inactive Yeast is a great protein source, with good amino acids balance and sufficient levels of Lysine, Methionine and Threonine and it is also the best source for natural occurring B vitamins.

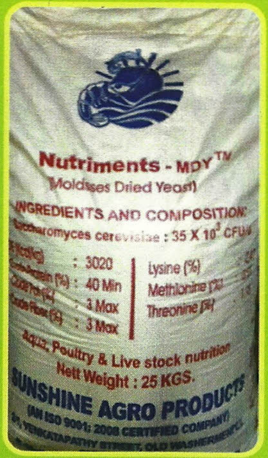
Microbial Study :

Saccharomyces cerevisiae : 35×10^3 cfu/g

Proximate composition (Typical) :

Parameters

Moisture (%)	8 Max
Ash (%)	8 Max
Crude Protein (%)	40 Min
Carbohydrate (%)	40
Ether extract (%)	0.02
Crude Fiber (%)	0.30
Gross energy swine (kcal/kg)	4690
ME Poultry (kcal/kg)	2218
ME Swine (kcal/kg)	3276
TDN (%)	78



Amino acids (Typical):

Amino Acids (%)	TAA	DAA-Swine	DAA-Poultry
Alanine	2.68	2.15	2.10
Arginine	1.75	1.58	1.22
Aspartic Acid	4.02	3.51	2.21
Glycine	1.65	1.32	1.22
Isoleucine	1.9	1.57	0.63
Leucine	2.58	2.17	0.96
Glutamic Acid	5.18	4.35	NE
Lysine	2.92	2.46	2.04
Cystine	0.29	0.21	0.16
Methionine	0.71	0.60	0.41
Phenylalanine	1.68	1.42	0.72
Tyrosine	1.09	0.95	0.80
Threonine	2.29	1.82	1.14
Tryptophan	0.45	NE	NE
Proline	1.30	NE	0.88
Valine	2.13	1.71	0.57
Histidine	1.10	0.94	0.59
Serine	2.18	0.94	1.040

TAA : Total Amino Acid   DAA : Digestible Amino Acid   NE : Not Evaluated